

Multiplying and Dividing Algebraic Fractions

Pre-Algebra Review

$$\textcircled{1} \frac{5x^4y^7}{7z^3} \left(-\frac{3x^3z}{10y^5}\right) = -\frac{15x^7y^2z}{70z^3y^5} = \boxed{-\frac{3x^7y^2}{14z^2}}$$

$$\textcircled{2} -\frac{2x^4y^7}{3z} \left(\frac{9xz^5}{11y^6}\right) = -\frac{18x^5y^7z^5}{33zy^6} = \boxed{-\frac{6x^5yz^4}{11}}$$

$$\textcircled{3} -\frac{4x^2y^2}{3z^3} (-12y^2) = \frac{48x^2y^4}{3z^3} = \boxed{\frac{16x^2y^4}{z^3}}$$

$$\textcircled{4} -7x^4y \left(\frac{3x}{14y}\right) = -\frac{21x^5y}{14y} = \boxed{-\frac{3x^5}{2}}$$

$$\textcircled{5} -\frac{3xy^7}{4z^5} \div \frac{3x^2z^5}{5y^3}$$

$$-\frac{3xy^7}{4z^5} \left(\frac{5y^3}{3x^2z^5}\right)$$

$$-\frac{15xy^{10}}{12x^2z^{10}}$$

$$\boxed{-\frac{5y^{10}}{4xz^{10}}}$$

$$\textcircled{6} \frac{2y^7}{5z^5} \div \left(-\frac{xz^3}{3y^3}\right)$$

$$\frac{2y^7}{5z^5} \left(-\frac{3y^3}{xz^3}\right)$$

$$\boxed{-\frac{6y^{10}}{5xz^8}}$$

$$\textcircled{7} -6x^8z^2 \div \left(-\frac{6x^5}{11y^4}\right)$$

$$\frac{-6x^8z^2}{1} \left(-\frac{11y^4}{6x^5}\right)$$

$$\frac{66x^3y^4z^2}{6x^5}$$

$$\boxed{11x^3y^4z^2}$$

$$\textcircled{8} -\frac{x^3y^7}{4z^5} \div (-4z^9x^5)$$

$$-\frac{x^3y^7}{4z^5} \left(-\frac{1}{4z^9x^5}\right)$$

$$\frac{x^3y^7}{16z^{14}x^5}$$

$$\boxed{\frac{y^7}{16z^{14}x^2}}$$

Beginning Algebra

$$\textcircled{9} \frac{x^2+9x+14}{3x^2+12x} \cdot \frac{x+4}{x^2-4}$$

$$\frac{(x+7)(x+2)}{3x(x+4)} \cdot \frac{x+4}{(x+2)(x-2)}$$

$$\boxed{\frac{x+7}{3x(x-2)}}$$

$$\textcircled{10} \frac{y^4}{y^2+4y-5} \cdot \frac{y^2+2y-15}{y^2}$$

$$\frac{y^4}{(y-1)(y+5)} \cdot \frac{(y+5)(y-3)}{y^2}$$

$$\boxed{\frac{y^2(y-3)}{y-1}}$$

$$\textcircled{11} \frac{z^2+11z+24}{z^2-7z+6} \cdot \frac{z^2-2z+1}{z^2+2z-3}$$

$$\frac{(z+3)(z+8)}{(z-6)(z-1)} \cdot \frac{(z-1)(z-1)}{(z+3)(z-1)}$$

$$\boxed{\frac{z+8}{z-6}}$$

$$\textcircled{12} \frac{a^2-81}{3a^2+23a+20} \cdot \frac{10a^5+10a^4}{a^2-a-72}$$

$$\frac{(a+9)(a-9)}{(3a+20)(a+1)} \cdot \frac{10a^4(a+1)}{(a-9)(a+8)}$$

$$\boxed{\frac{10a^4(a+9)}{(3a+20)(a+8)}}$$

$$\textcircled{13} \frac{b^2-81}{3b-5} \cdot \frac{9b^2-25}{b^2+14b+45}$$

$$\frac{(b-9)(b+9)}{3b-5} \cdot \frac{(3b-5)(3b+5)}{(b+5)(b+9)}$$

$$\boxed{\frac{(b-9)(3b+5)}{b+5}}$$

$$\textcircled{14} \frac{x^4-1}{6x^3-8x} \cdot \frac{27x^5-36x^3}{x^4+2x^2+1}$$

$$\frac{(x^2+1)(x^2-1)}{2x(3x^2-4)} \cdot \frac{9x^3(3x^2-4)}{(x^2+1)(x^2+1)}$$

$$\boxed{\frac{9x^2(x^2-1)}{2(x^2+1)}}$$

$$\textcircled{15} \frac{x^2+x-12}{x-2} \cdot \frac{3x^2-6x}{x^2-9}$$

$$\frac{(x+4)(x-3)}{x-2} \cdot \frac{3x(x-2)}{(x-3)(x+3)}$$

$$\boxed{\frac{3x(x+4)}{x+3}}$$

$$\textcircled{16} \frac{c^2}{c+5} \cdot \frac{c^2+10c+25}{c^3-3c^2-40c}$$

$$\frac{c^2}{c+5} \cdot \frac{(c+5)(c+5)}{c(c^2-3c-40)}$$

$$\frac{c^2}{c+5} \cdot \frac{(c+5)(c+5)}{c(c+5)(c-8)}$$

$$\boxed{\frac{c}{c-8}}$$

$$\textcircled{17} \frac{z^2+8z+12}{z^2+10z+24} \cdot \frac{z^2+3z-4}{z^2-14z+13}$$

$$\frac{(z+2)(z+6)}{(z+4)(z+6)} \cdot \frac{(z-1)(z+4)}{(z-13)(z-1)}$$

$$\boxed{\frac{z+2}{z-13}}$$

$$\textcircled{18} \frac{t^3-49t}{6t^3-42t^2} \cdot \frac{8t}{4t^3+28t^2}$$

$$\frac{t(t^2-49)}{6t^2(t-7)} \cdot \frac{8t}{4t^2(t+7)}$$

$$\frac{t(t-7)(t+7)}{6t^2(t-7)} \cdot \frac{8t}{4t^2(t+7)}$$

$$\frac{8t^2}{24t^4} = \boxed{\frac{1}{3t^2}}$$

$$\textcircled{19} \frac{2a^2+17a+30}{a^2-12a+20} \cdot \frac{a^2-4}{-5a^2-30a}$$

$$\frac{(2a+5)(a+6)}{(a-2)(a-10)} \cdot \frac{(a-2)(a+2)}{-5a(a+6)}$$

$$\boxed{\frac{(2a+5)(a+2)}{-5a(a-10)}}$$

$$\textcircled{20} \frac{25b^2-64}{b^2-36} \cdot \frac{b+6}{5b^2+2b-16}$$

$$\frac{(5b-8)(5b+8)}{(b+6)(b-6)} \cdot \frac{b+6}{(5b-8)(b+2)}$$

$$\boxed{\frac{5b+8}{(b-6)(b+2)}}$$

$$\textcircled{21} \frac{y-4}{y^2-6y+9} \div \frac{y^2+y-20}{y^2-9}$$

$$\frac{y-4}{y^2-6y+9} \cdot \frac{y^2-9}{y^2+y-20}$$

$$\frac{y-4}{(y-3)(y-3)} \cdot \frac{(y-3)(y+3)}{(y+5)(y-4)}$$

$$\boxed{\frac{y+3}{(y-3)(y+5)}}$$

$$\textcircled{22} \frac{22z^3+11z^2}{z^2-7z} \div \frac{z^2+3z+2}{z^2-5z-14}$$

$$\frac{22z^3+11z^2}{z^2-7z} \cdot \frac{z^2-5z-14}{z^2+3z+2}$$

$$\frac{11z^2(2z+1)}{z(z-7)} \cdot \frac{(z-7)(z+2)}{(z+1)(z+2)}$$

$$\boxed{\frac{11z(2z+1)}{z+1}}$$

$$(23) \frac{a^2-6a-16}{49a^2-36} \div \frac{a-8}{7a+6}$$

$$\frac{a^2-6a-16}{49a^2-36} \cdot \frac{7a+6}{a-8}$$

$$\frac{(a-8)(a+2)}{(7a-6)(7a+6)} \cdot \frac{7a+6}{a-8}$$

$$\boxed{\frac{a+2}{7a-6}}$$

$$(24) \frac{b^2-16}{b^2-6b+8} \div \frac{b^2-8b-48}{b^2-17b+30}$$

$$\frac{b^2-16}{b^2-6b+8} \cdot \frac{b^2-17b+30}{b^2-8b-48}$$

$$\frac{(b-4)(b+4)}{(b-4)(b-2)} \cdot \frac{(b-2)(b-15)}{(b-12)(b+4)}$$

$$\boxed{\frac{b-15}{b-12}}$$

$$(27) \frac{7z^2+23z+6}{z^2-7z-18} \div \frac{49z^2-4}{z^2-81}$$

$$\frac{7z^2+23z+6}{z^2-7z-18} \cdot \frac{z^2-81}{49z^2-4}$$

$$\frac{(7z+2)(z+3)}{(z-9)(z+2)} \cdot \frac{(z-9)(z+9)}{(7z+2)(7z-2)}$$

$$\boxed{\frac{(z+3)(z+9)}{(z+2)(7z-2)}}$$

$$(28) \frac{a-1}{a^2} \div \frac{a^2-1}{a+1}$$

$$\frac{a-1}{a^2} \cdot \frac{a+1}{a^2-1}$$

$$\frac{a-1}{a^2} \cdot \frac{a+1}{(a-1)(a+1)}$$

$$\boxed{\frac{1}{a^2}}$$

-1 Rule

$$(25) \frac{5}{3-2z} \div \frac{10z^2+11z-6}{4z^2-9}$$

$$\frac{5}{3-2z} \cdot \frac{4z^2-9}{10z^2+11z-6}$$

$$\frac{5}{-1(-3+2z)} \cdot \frac{(2z-3)(2z+3)}{(2z+3)(5z-2)}$$

$$\boxed{-\frac{5}{5z-2}}$$

$$(26) \frac{y^2-4}{y^2-15y+54} \div \frac{y^2-18y-40}{y-9}$$

$$\frac{y^2-4}{y^2-15y+54} \cdot \frac{y-9}{y^2-18y-40}$$

$$\frac{(y+2)(y-2)}{(y-6)(y-9)} \cdot \frac{y-9}{(y-20)(y+2)}$$

$$\boxed{\frac{y-2}{(y-6)(y-20)}}$$

$$(29) \frac{10b^2-1000}{3b^5-42b^4+99b^3} \div \frac{2b^2-4b-160}{4b^2-4b-24}$$

$$\frac{10b^2-1000}{3b^5-42b^4+99b^3} \cdot \frac{4b^2-4b-24}{2b^2-4b-160}$$

$$\frac{10(b^2-100)}{3b^3(b^2-14b+33)} \cdot \frac{4(b^2-b-6)}{2(b^2-2b-80)}$$

$$\frac{10(b-10)(b+10)}{3b^3(b-3)(b-11)} \cdot \frac{4(b-3)(b+2)}{2(b-10)(b+8)}$$

$$\boxed{\frac{20(b+10)(b+2)}{3b^3(b-11)(b+8)}}$$

$$(30) \frac{-2}{3z+8} \div \frac{8-3z}{9z^2-64}$$

$$\frac{-2}{3z+8} \cdot \frac{9z^2-64}{8-3z}$$

$$\frac{-2}{3z+8} \cdot \frac{(3z-8)(3z+8)}{-1(-8+3z)}$$

$$\boxed{2}$$

$$\textcircled{31} \frac{yx^2 - 10xy}{x^2 - 3xy + 2y^2} \div \frac{x^2 - 100}{x^2 - y^2}$$

$$\frac{yx^2 - 10xy}{x^2 - 3xy + 2y^2} \cdot \frac{x^2 - y^2}{x^2 - 100}$$

$$\frac{yx(x-10)}{(x-2y)(x-y)} \cdot \frac{(x-y)(x+y)}{(x-10)(x+10)}$$

$$\boxed{\frac{yx(x+y)}{(x-2y)(x+10)}}$$

$$\textcircled{32} \frac{c^2 - 9d^2}{2c^2 - cd - d^2} \div \frac{c^3 + 3dc^2}{c^2 - d^2}$$

$$\frac{c^2 - 9d^2}{2c^2 - cd - d^2} \cdot \frac{c^2 - d^2}{c^3 + 3dc^2}$$

$$\frac{(c-3d)(c+3d)}{(2c+d)(c-d)} \cdot \frac{(c-d)(c+d)}{c^2(c+3d)}$$

$$\boxed{\frac{(c-3d)(c+d)}{c^2(2c+d)}}$$

$$\textcircled{33} \frac{4a^2}{a^2 + 45a - 250} \div \frac{2a^2 - 5a}{25 - a^2}$$

$$\frac{4a^2}{a^2 + 45a - 250} \cdot \frac{25 - a^2}{2a^2 - 5a}$$

$$\frac{4a^2}{(a+50)(a-5)} \cdot \frac{-1(-25+a^2)}{a(2a-5)}$$

$$\frac{4a^2}{(a+50)(a-5)} \cdot \frac{-1(a^2-25)}{a(2a-5)}$$

$$\frac{4a^2}{(a+50)(a-5)} \cdot \frac{-1(a-5)(a+5)}{a(2a-5)}$$

$$\boxed{\frac{-4a(a+5)}{(a+50)(2a-5)}}$$

$$\textcircled{34} \frac{4a^6}{1 - 4a^2} \div \frac{2a^2 + 12a + 10}{2a - 1}$$

$$\frac{4a^6}{1 - 4a^2} \cdot \frac{2a - 1}{2a^2 + 12a + 10}$$

$$\frac{4a^6}{-1(-1+4a^2)} \cdot \frac{2a-1}{2a^2+12a+10}$$

$$\frac{4a^6}{-1(4a^2-1)} \cdot \frac{2a-1}{2(a^2+6a+5)}$$

$$\frac{4a^6}{-1(2a-1)(2a+1)} \cdot \frac{2a-1}{2(a+5)(a+1)}$$

$$\boxed{\frac{2a^6}{(2a+1)(a+5)(a+1)}}$$

$$\textcircled{35} \frac{b-3}{b^4 - 10b^2 + 9} \div \frac{b^4}{-2b^6 - 6b^5}$$

$$\frac{b-3}{b^4 - 10b^2 + 9} \cdot \frac{-2b^6 - 6b^5}{b^4}$$

$$\frac{b-3}{(b^2-9)(b^2-1)} \cdot \frac{-2b^5(b+3)}{b^4}$$

$$\frac{b-3}{(b-3)(b+3)(b^2-1)} \cdot \frac{-2b^5(b+3)}{b^4}$$

$$\boxed{\frac{2b}{b^2-1}}$$

$$\textcircled{36} \frac{b-9}{2b^4 - 3b^3 - 5b^2} \div \frac{-b^2 + 13b - 36}{-12b^6 + 75b^4}$$

$$\frac{b-9}{2b^4 - 3b^3 - 5b^2} \cdot \frac{-12b^6 + 75b^4}{-b^2 + 13b - 36}$$

$$\frac{b-9}{b^2(2b^2-3b-5)} \cdot \frac{-3b^4(4b^2-25)}{-1(b^2-13b+36)}$$

$$\frac{b-9}{b^2(2b-5)(b+1)} \cdot \frac{-3b^4(2b-5)(2b+5)}{-1(b-4)(b-9)}$$

$$\boxed{\frac{3b^2(2b+5)}{(b+1)(b-4)}}$$

$$\textcircled{37} \frac{x^7 + 3x^6}{-9 + x^2} \cdot \frac{x+1}{-x^2 + 4x + 32} \cdot \frac{1}{x^2 - 1} \cdot \frac{x+4}{6x^4 - 5x^3}$$

$$\frac{x^6(x+3)}{x^2-9} \cdot \frac{x+1}{-1(x^2-4x-32)} \cdot \frac{1}{(x-1)(x+1)} \cdot \frac{x+4}{x^3(6x-5)}$$

$$\frac{x^6(x+3)}{(x-3)(x+3)} \cdot \frac{x+1}{-1(x-8)(x+4)} \cdot \frac{1}{(x-1)(x+1)} \cdot \frac{x+4}{x^3(6x-5)}$$

$$\boxed{\frac{x^3}{(x-3)(x-8)(x-1)(6x-5)}}$$

$$(38) \frac{x^4 - 15x^2 - 16}{9x^2 + 36x} \cdot \frac{x+4}{x^2 + 15x + 56} \cdot \frac{3x^2}{20 - 5x} \cdot \frac{-49 + x^2}{x - 9}$$

$$\frac{(x^2 - 16)(x^2 + 1)}{9x(x+4)} \cdot \frac{x+4}{(x+7)(x+8)} \cdot \frac{3x^2}{-5(-4+x)} \cdot \frac{x^2 - 49}{x - 9}$$

$$\frac{(x-4)(x+4)(x^2+1)}{9x(x+4)} \cdot \frac{x+4}{(x+7)(x+8)} \cdot \frac{3x^2}{-5(-4+x)} \cdot \frac{(x-7)(x+7)}{x-9}$$

$$= \frac{x(x^2+1)(x+4)(x-7)}{15(x+8)(x-9)}$$